DATA ENGINEERING

Data Quality Visual Guide

Trust your tables.
Checks, anomalies, SLAs, and quarantine patterns.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

The Six Dimensions of Quality

If you can't name the dimension, you can't measure it.

Accuracy
Values reflect reality

Completeness
No missing critical fields

Consistency
Same fact, same value everywhere

Timeliness
Fresh enough for the decision

Validity
Conforms to format/domain rules

Uniqueness
No unintended duplicates

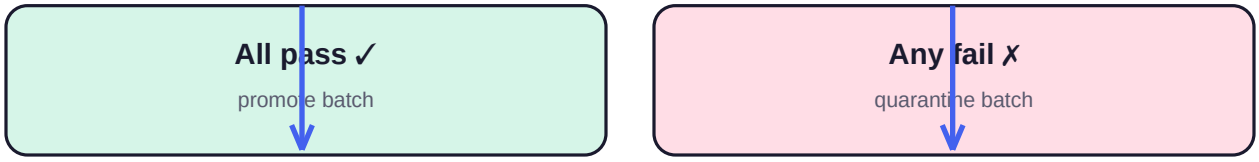
Every check you write maps to one of these. A 'data quality program' is just these six, measured continuously.

STEP 2/6

Expectation Checks

Declarative assertions on every batch — the Great Expectations pattern.

```
expect_column_values_to_not_be_null("user_id")
expect_column_values_to_be_between("age", 0, 120)
expect_column_values_to_be_unique("order_id")
expect_table_row_count_to_be_between(9000, 11000)
```

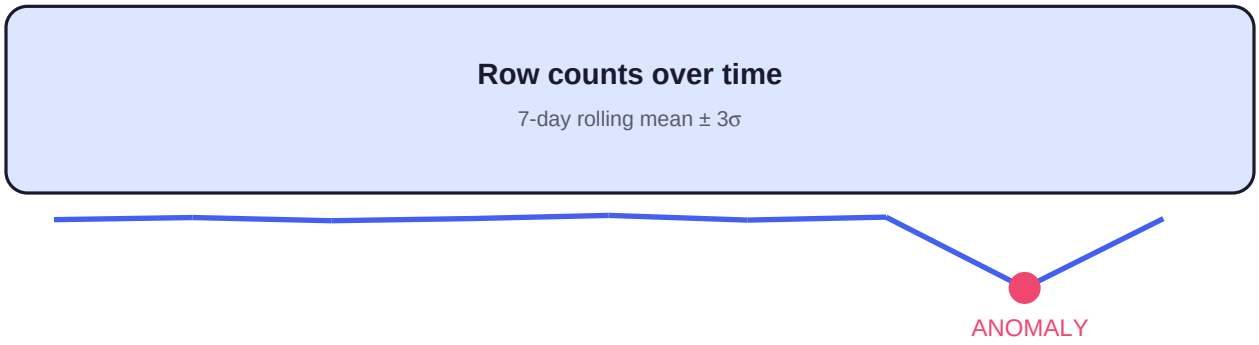


Expectations run as part of the pipeline (dbt tests, Soda, GE). A failing suite blocks promotion — quality as a gate, not a report.

STEP 3/6

Anomaly Detection

Rules catch known bad; statistics catch unknown bad.



Track volume, null rates, and distribution drift per table. A sudden dip often means an upstream source broke — hours before anyone notices.

- Freshness monitors: alert when a table misses its SLA
- Schema drift detection: new/removed columns break silent contracts
- Seasonality-aware baselines beat fixed thresholds

STEP 4/6

Data SLAs

A contract between producers and consumers: what, when, how good.

orders table SLA

- Freshness: landed by 06:00 ET daily
- Completeness: ≥ 99.5% of source rows
- Validity: 100% of amounts ≥ 0, ISO currency codes

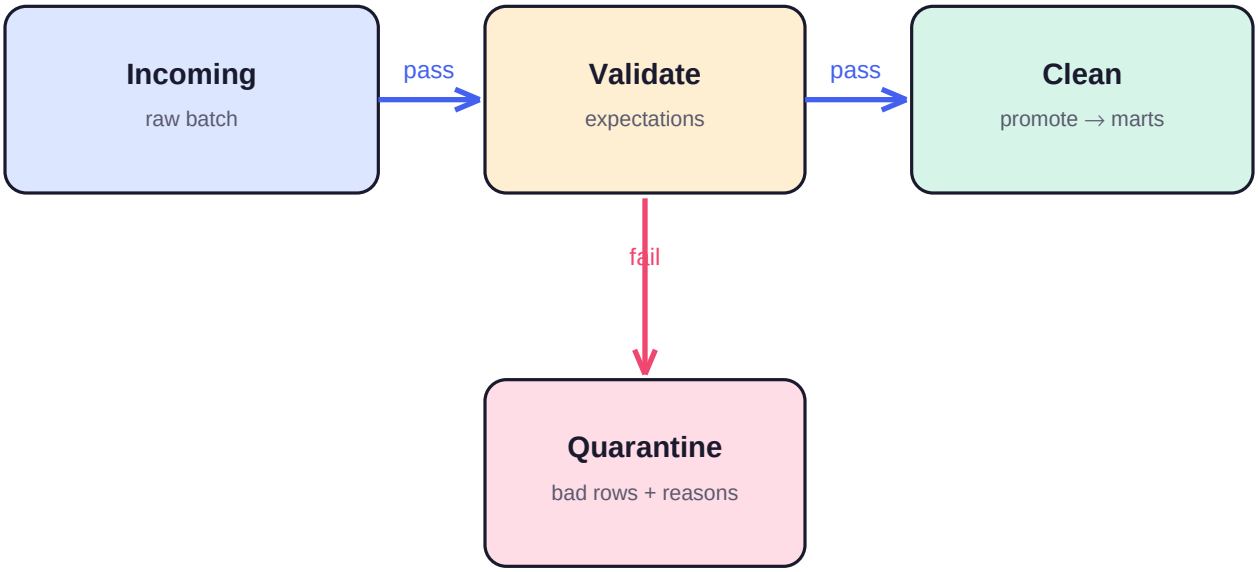
SLAs turn vague complaints ('the data feels off') into measurable breaches with owners and escalation paths.

- Each SLA names an owner — unowned SLAs are wishes
- Breach → incident: page the producer, notify consumers
- Review quarterly: stale SLAs erode trust in all of them

STEP 5/6

Quarantine Pattern

Bad rows go somewhere — not into production, not into the void.

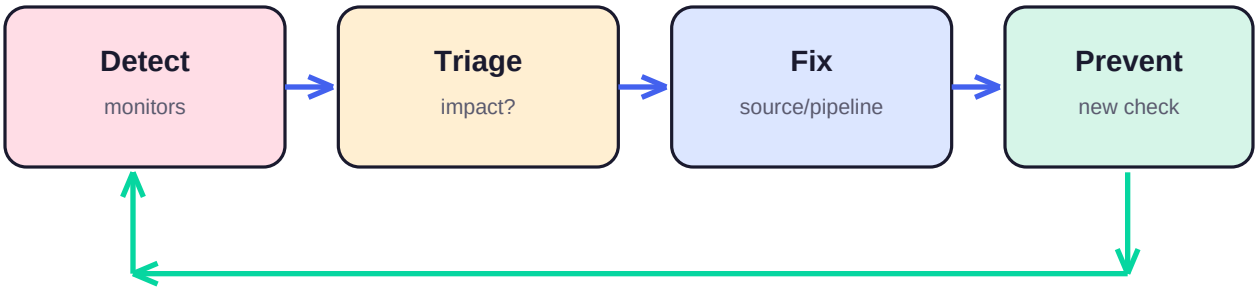


Quarantine tables keep the evidence: the row, the failed rule, the timestamp. Producers fix the source; analysts see what's excluded and why.

STEP 6/6

The Quality Loop

Detect → triage → fix → prevent. Quality compounds.



Every incident should produce a new automated check. A quality program that doesn't grow checks from incidents is just monitoring theater.

STEP ✓I✓

Key Takeaways

Data Quality — Remember

- Measure the six dimensions: accuracy, completeness, consistency, timeliness, validity, uniqueness.
- Expectation checks (dbt tests, Soda, GE) gate promotion — bad data fails the build.
- Anomaly detection on volume, null rates, and drift catches unknown-unknowns.
- Data SLAs make quality contractual: named owners, measurable targets, breach escalation.
- Quarantine bad rows with reasons attached; every incident should birth a new check.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev